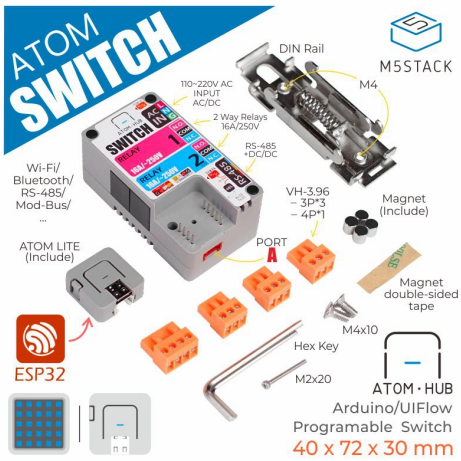
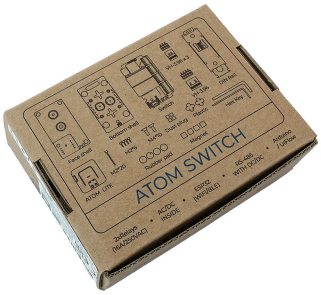
Atom Switch

SKU:K042



















Description

Atom Switch is a dual-channel programmable switch that supports ATOM, allowing simultaneous control of two AC/DC circuits. It features a built-in step-down module that provides 5V/1A DC power to the ATOM. The onboard relay supports up to 250V/10A mains (16A instantaneous current). To ensure safety, the DC circuit includes overheat and short-circuit protection, effectively disconnecting the circuit in case of excessive current or short-circuit. For industrial applications, we have integrated an SP485EE level conversion chip, providing an RS485 interface for connecting RS485 devices. The RS485 interface can also power the ATOM, supporting a voltage range of 9-24V.

Additionally, a HY2.0 interface is provided for connecting I2C peripherals or general I/O devices. Leveraging the ATOM's WIFI capabilities, you can easily achieve remote control of the switch. If you have multiple **Atom Switch** units, they can be connected in parallel via the RS485 interface.

Features

- Built-in AC-DC power supply circuit
- 2-channel relay
- Built-in RS485 level conversion, supports Modbus
- HY2.0 expansion
- Short-circuit and overheat protection
- Remote control via WIFI and RS485

Includes

- 1 x Atom-Lite
- 1 x Atom Switch
- 1 x Double-sided tape
- 1 x Rail clamp
- 1 x HT3.96-4P terminal
- 3 x 3.96 x 3P male headers
- 1 x M4 hex key
- 1 x M2 hex key
- 2 x M4*10mm hex socket head screws
- 1 x M2*20mm hex socket head machine screw
- 1 x I/O Sticker

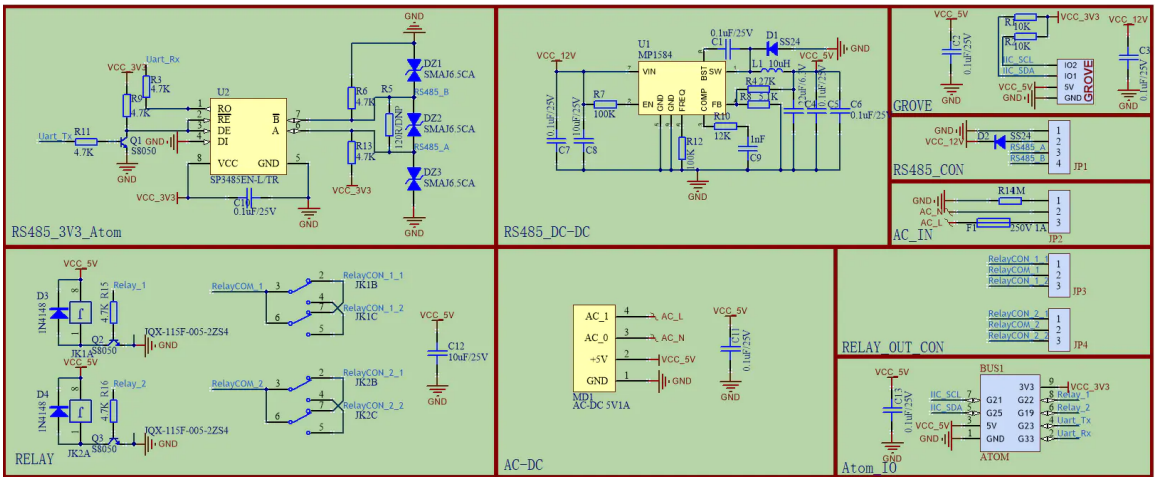
Applications

- Smart switches

Specifications

Specification	Parameters
Relay Parameters	AC 250V/10A (16A instantaneous)
Switch Power (AC-DC)	AC 250V-DC 5V
RS485 Power Supply	DC 9 ~ 24V
Interfaces	1 x HY2.0-4P (PORT.A), 1 x HT3.96-4P (RS485), 2 x HT3.96-2P (Relay), 1 x HT3.96-2P (AC/DC IN)
Housing Material	Plastic (PC)
Product Size	72 x 40 x 30mm
Product Weight	80.0g
Package Size	150.0 x 77.0 x 35.0mm
Gross Weight	168.4g

Schematics

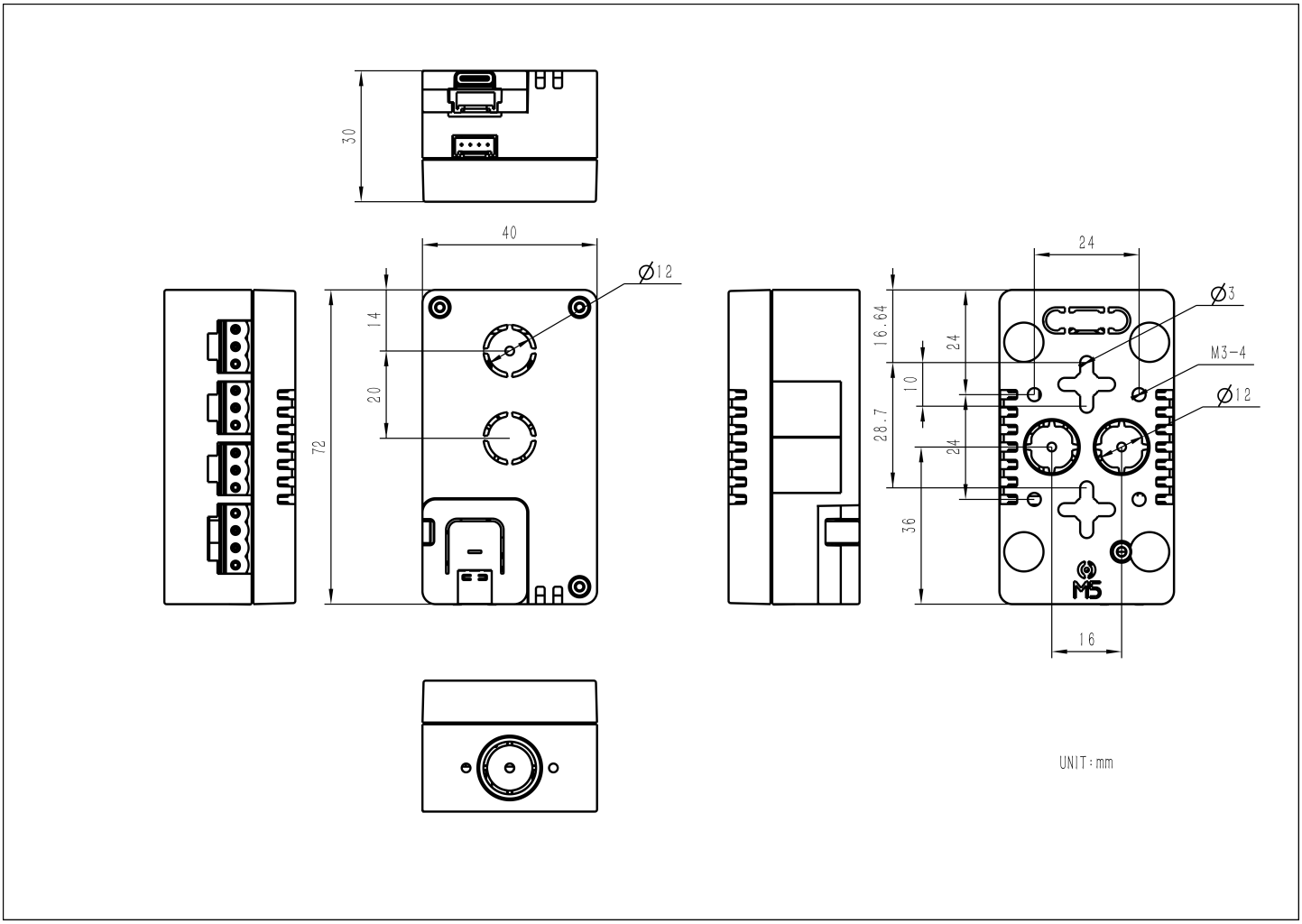


PinMap

PIN	LEFT	RIGHT	PIN
		1	3V3
PORT.A	2	3	RELAY1
PORT.A	4	5	RELAY2
5V	6	7	RS485_RX
GND	8	9	RS485_TX

Model Size

- Atom Switch Model Size PDF



Datasheets

- SP485EE

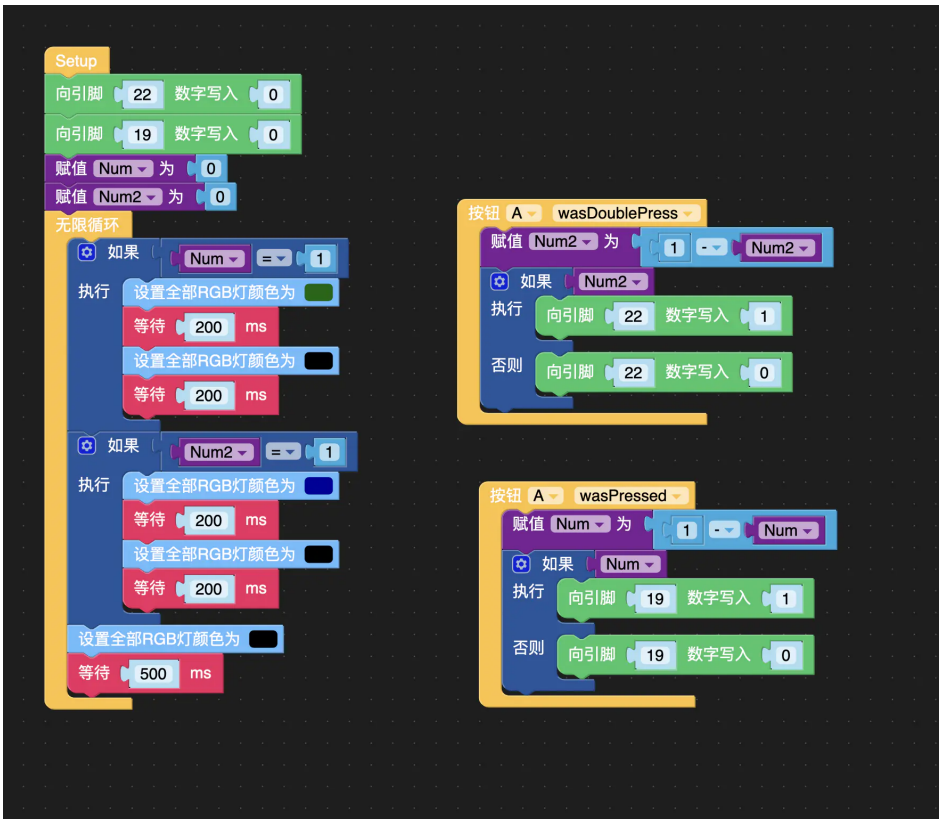
Softwares

Arduino

- Atom Switch Test Example

UiFlow1

- Atom Switch Test Example



EasyLoader

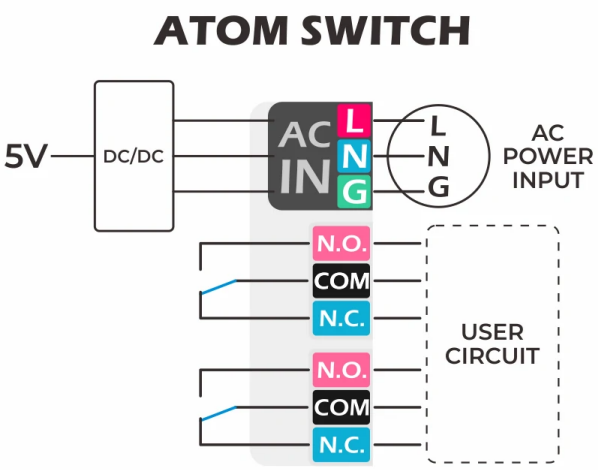
Easyloader	Download Link	Notes
Atom Switch Easyloader	download	/

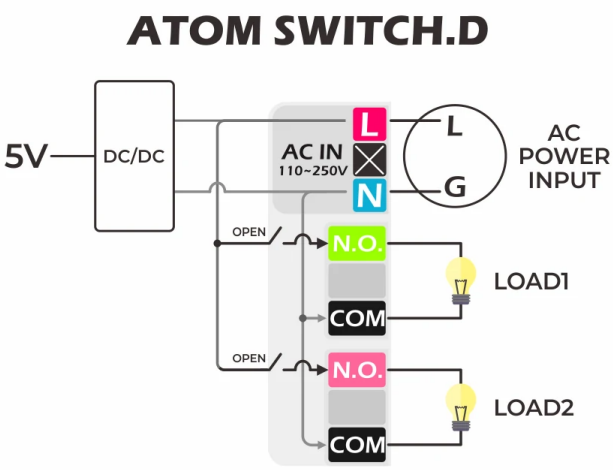
Video

AtomSwitch.mp4

Switch & Switch-D Comparison

/	Switch	Switch-D
Relay Current	AC 250V/10A	AC 250V/5A
Power Input	HT3.96R 2P	HT3.96R 3P
Relay Interface	(NO,NC,COM) x2	(NO (directly connected to AC L), COM (directly connected to AC N)) x2





Learn

The kit includes rail mounting hardware. Please refer to the following images for installation instructions:

